

The Mediterranean diet improves the systemic lipid and DNA oxidative damage in metabolic syndrome individuals. A randomized, controlled, trial.



BACKGROUND & AIMS: Metabolic syndrome (MetS), in which a non-classic feature is an increase in systemic oxidative biomarkers, presents a high risk of diabetes and cardiovascular disease (CVD). Adherence to the Mediterranean Diet (MedDiet) is associated with a reduced risk of MetS. However, the effect of the MedDiet on biomarkers for oxidative damage has not been assessed in MetS individuals. We have investigated the effect of the MedDiet on systemic oxidative biomarkers in MetS individuals. METHODS: Randomized, controlled, parallel clinical trial in which 110 female with MetS, aged 55-80, were recruited into a large trial (PREDIMED Study) to test the efficacy of the traditional MedDiet on the primary prevention of CVD. Participants were assigned to a low-fat diet or two traditional MedDiets (MedDiet + virgin olive oil or MedDiet + nuts). Both MedDiet group participants received nutritional education and either free extra virgin olive oil for all the family (1 L/week), or free nuts (30 g/day). Diets were ad libitum. Changes in urine levels of F2-Isoprostane (F2-IP) and the DNA damage base 8-oxo-7,8-dihydro-2'-deoxyguanosine (8-oxo-dG) were evaluated at 1-year trial. RESULTS: After 1-year urinary F2-IP decreased in all groups, the decrease in MedDiet groups reaching a borderline significance versus that of the Control group. Urinary 8-oxo-dG was also reduced in all groups, with a higher decrease in both MedDiet groups versus the Control one ($P < 0.001$). CONCLUSIONS: MedDiet reduces oxidative damage to lipids and DNA in MetS individuals. Data from this study provide evidence to recommend the traditional MedDiet as a useful tool in the MetS management. Registered under Clinical Trials.gov Identifier no. NCT00123456. Copyright © 2012 Elsevier Ltd and European Society for Clinical Nutrition and Metabolism. All rights reserved.